

Deducing climate history

CH. 2.3–2.4

How is climate history deduced ?

- **Paleoproxies:** Long-lived geological, chemical, or biological systems that have the climate imprinted on them.
 - E.g. Ice sheets
 - E.g. Tree rings
 - E.g. Ocean sediments
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- Putting all of these **paleoproxies** together gives us a reasonably complete picture of the global climate going back hundreds of millions of years, with some information about the climate going back billions of years.

Ice cores

- Chemical composition of the ice, **size and orientation of the ice crystals** → air temperature.
- Small air bubbles trapped → chemical composition of the atmosphere.
- Dust trapped in the ice → prevailing wind speed and direction. More dust blows around during droughts, giving information about how wet or dry the regional climate was when the ice formed.
- Sulfur in glacial ice show → major volcanic eruption.
- Age of each ice layer → depth inside the glacier.
- Ice cores only provide climatic information in regions and over time periods that are cold enough for permanent ice to exist. E.g. Greenland, Antarctica, and glaciers. Ice cores from the thickest, oldest ice in Antarctica have provided information about the climate going back an amazing **2 million years.**

Tree rings

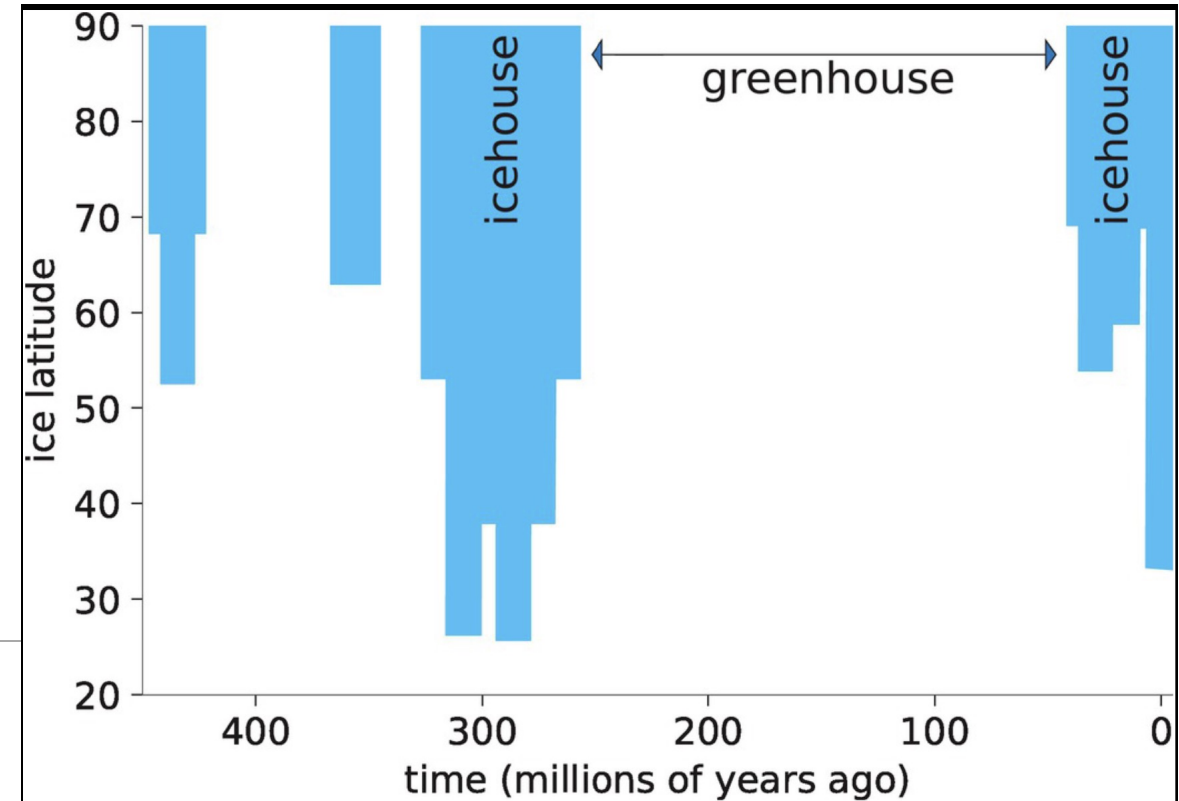
- Trees store climate information in their tree rings.
- Tree growth follows an annual cycle, which is imprinted in the rings in their trunks.
- Spring → Rapid trees growth → light-colored wood
- Fall → slow growth → produce dark wood.
- **Warm and wet years** → more tree growth → produces wider rings
- Width of each ring → climate conditions around that tree in that year.
- Tree rings only reveal information about the climate as far back as wood from them is available i.e. for the **past millennium** or so.

Ocean sediments

- Ocean sediments accumulate at the bottom of the ocean every year → climate conditions at the time they were deposited.
- Type of skeleton of marine organisms → surface water temperature
- Chemical composition of the skeletons and variations in size and shape of particular species → water temperature, salinity, dissolved oxygen, atmospheric CO₂, nearby continental precipitation, the strength and direction of the prevailing winds, and nutrient availability
- This information goes back **tens of millions of years.**

Earth's long-term temperature

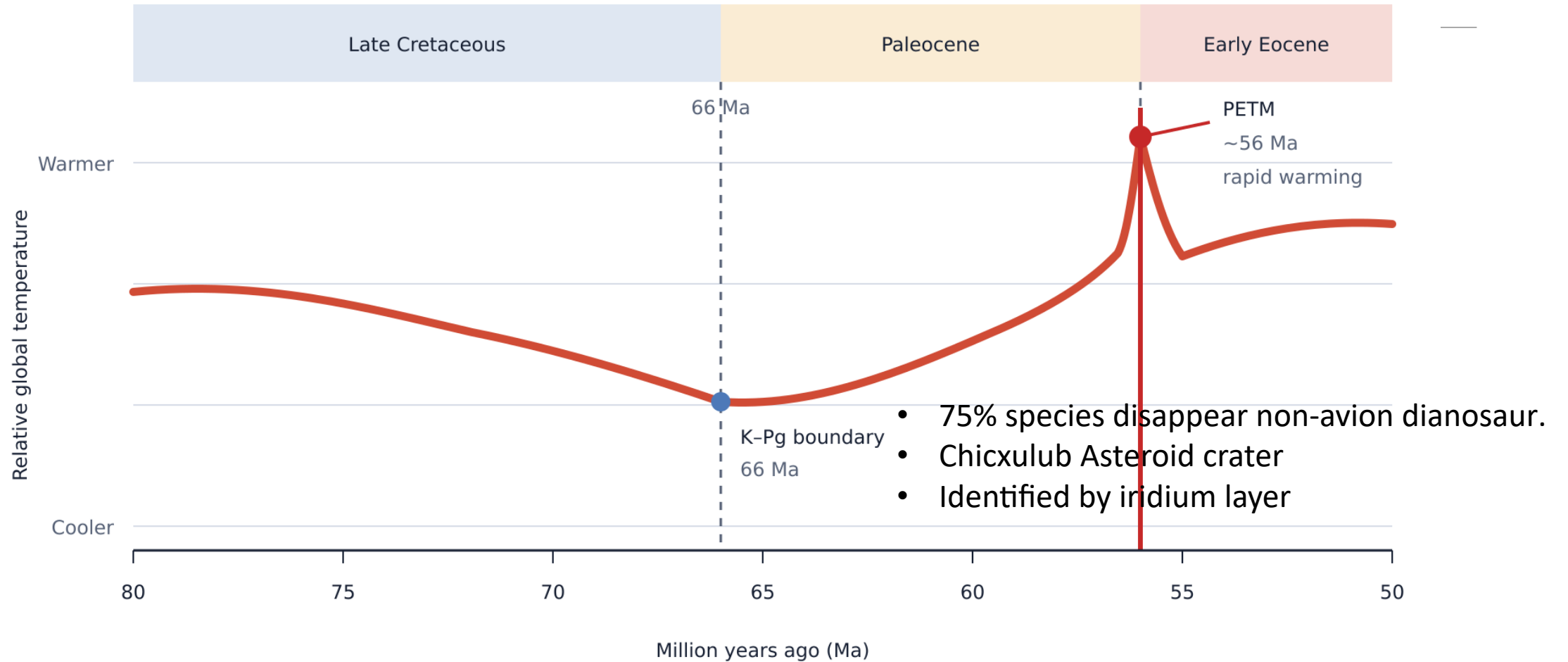
- Earth's climate has not been constant — it has shifted between very different climate states- icehouse and greenhouse periods
- E.g. Sedimentary rocks 4 billion years ago (indicating liquid water) and ice sheets 700 million years ago (snowball earth)



Extent of continental ice deposits, showing icehouse-greenhouse cycles. The blue bars show the largest latitudinal extent of ice over time.

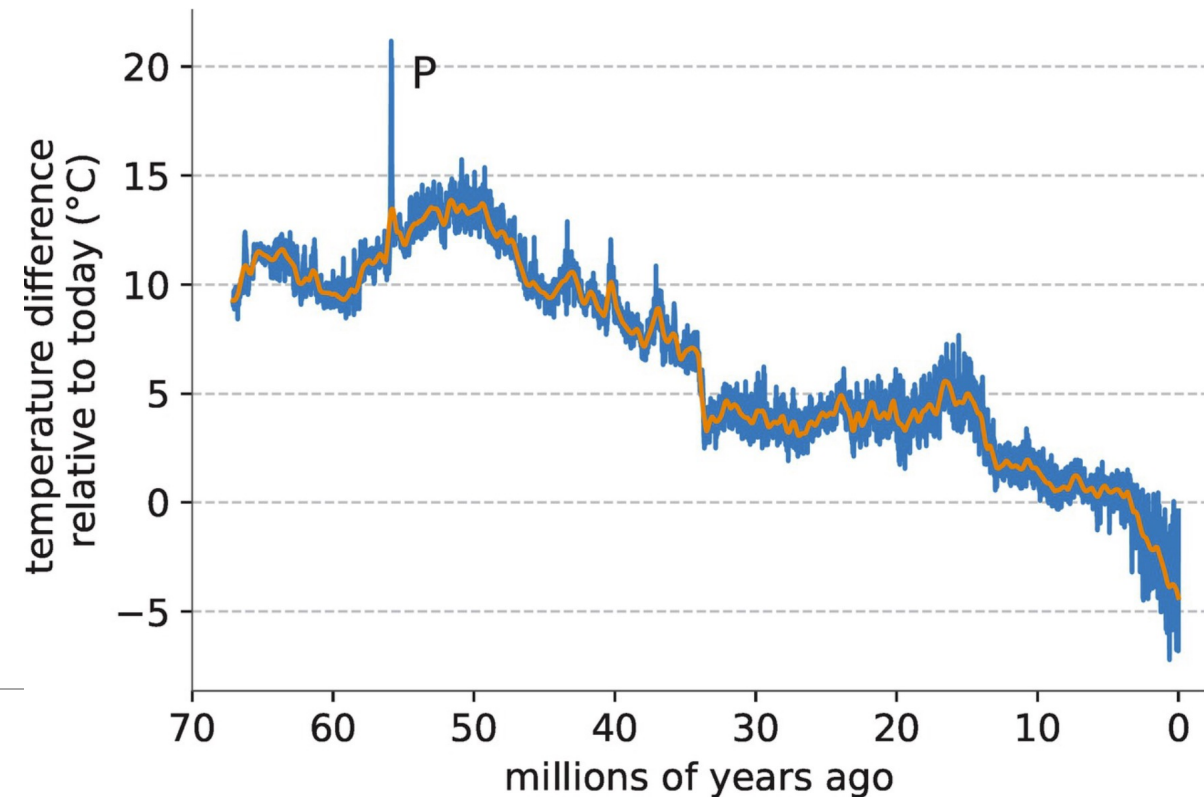
Climate history 80 M year to 50 M year

Climate history: 80–50 million years ago
Schematic global temperature trend (older → younger)



Paleocene–Eocene Thermal Maximum

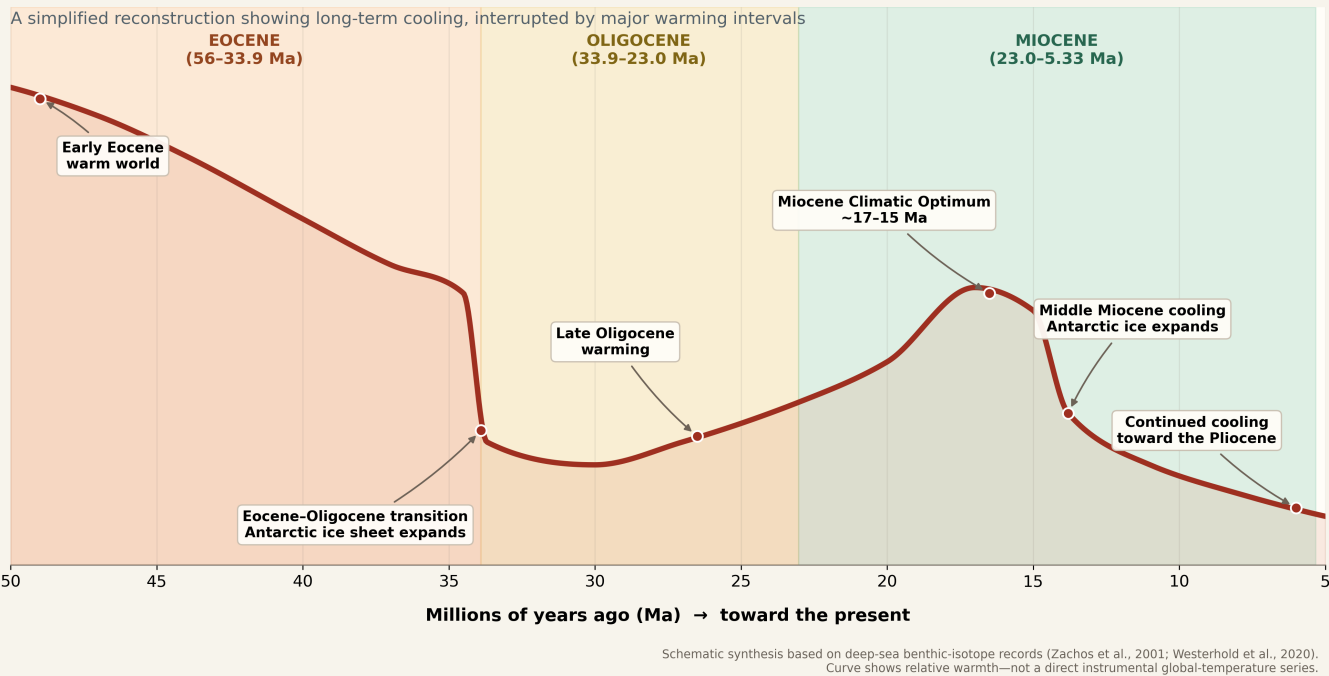
- 55 million years ago: **Paleocene-Eocene Thermal Maximum (PETM)** abrupt warming of more than 5°C that occurred over a few 1000 years due to massive release of GHGs (similar to today). (Jiang et al 2025, Nature Communication)
- 50 million year ago: warmest temp. occurred in a period called **Eocene Climatic Optimum**.
- 35 million years ago: dinosaurs extinct
- Later, pre-PETM temperatures were restored.



Reconstructed global average surface temperature over the past 70 million years, relative to today's temperature. The sharp temperature spike 55 million years ago (labeled "P") represents the Paleocene—Eocene Thermal Maximum (PETM).

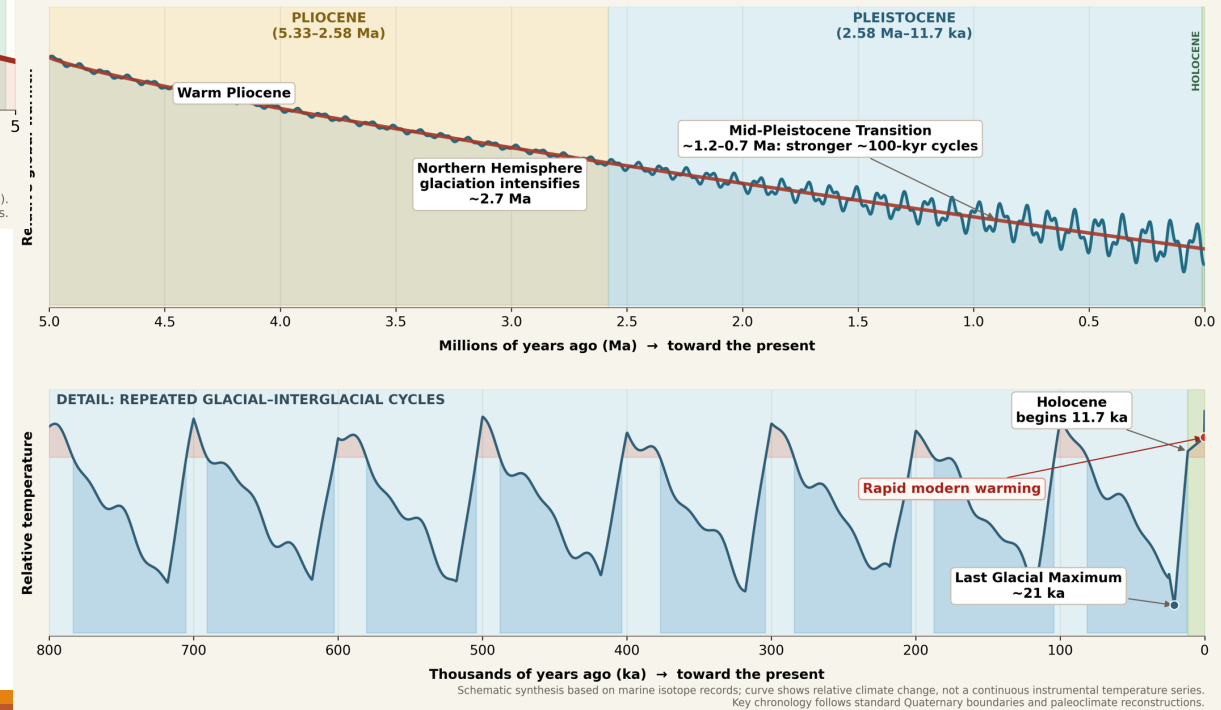
Climate change history

Earth's Climate History: 50-5 Million Years Ago



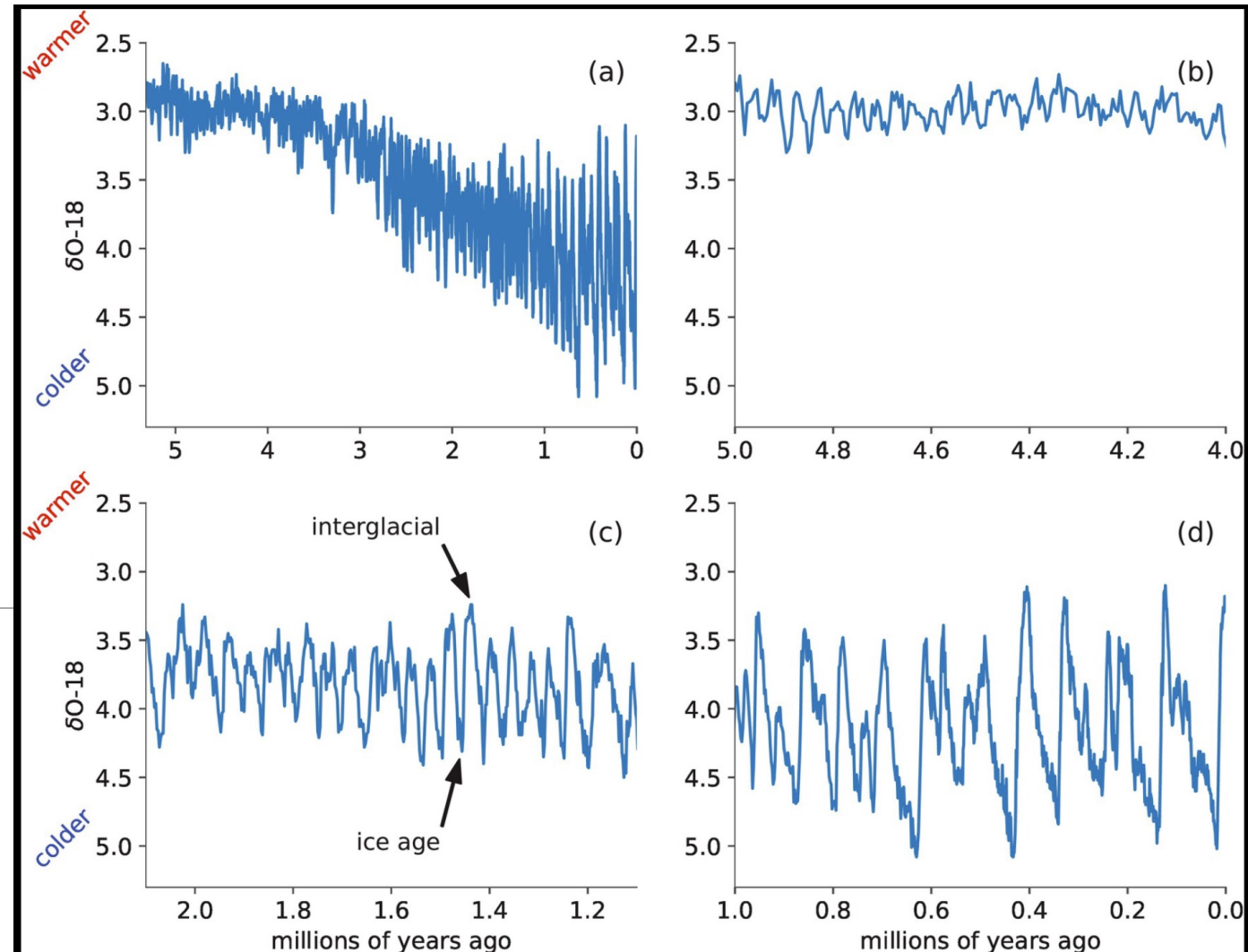
Earth's Climate History: 5 Million Years Ago to the Present

Long-term cooling led to Northern Hemisphere ice ages, followed by repeated glacial cycles and recent rapid warming



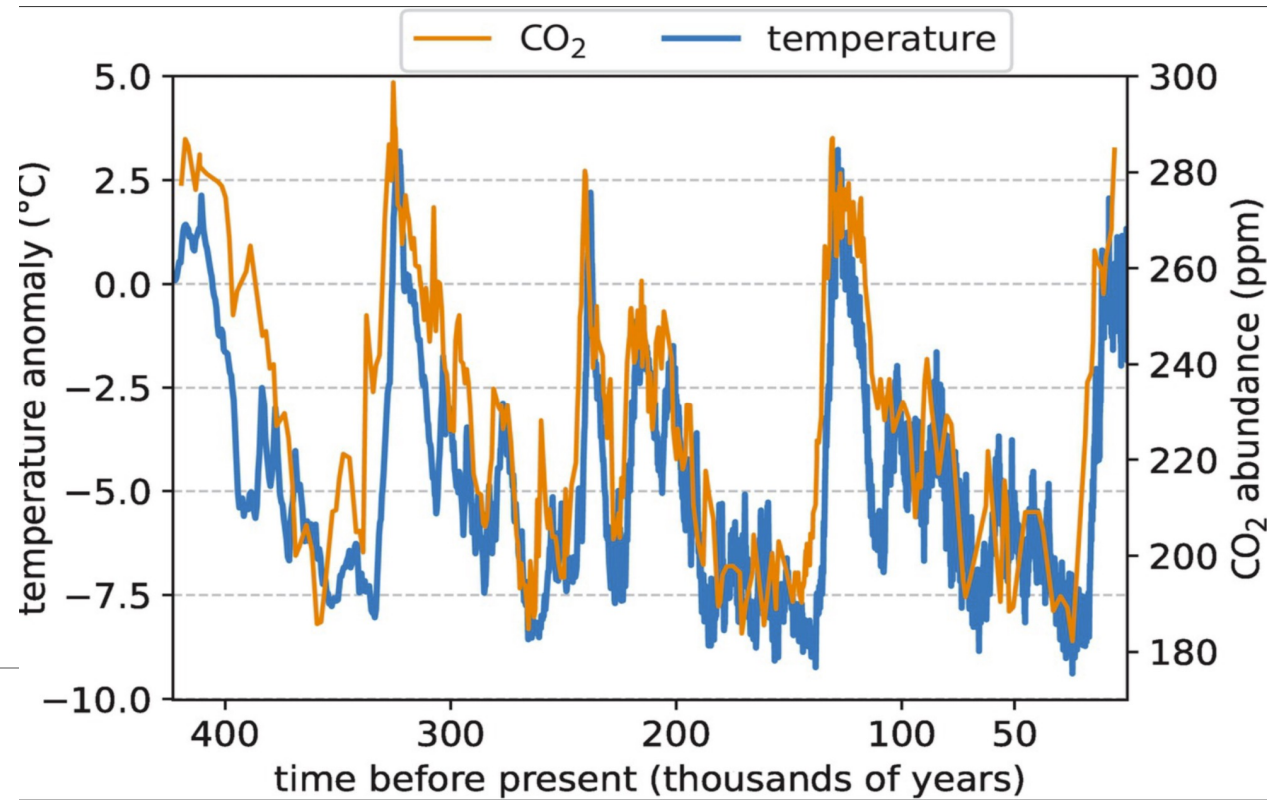
Global temp. variations 5.3 M years ago

- **Ice ages** are cool periods when ice sheets expanded to cover large parts of the northern hemisphere's land areas.
- During the warm periods between the ice ages, called **interglacials**, the ice sheets contracted.
- ❖ The vertical axis measures the relative abundance of oxygen-18, a heavy isotope of oxygen that is a proxy for temperature, in the ice and ocean sediment cores.
- ❖ For ice core the ^{18}O depletion per degree C is 0.6-0.7 mil, while for marine carbonates it was found 0.22-0.25 mil.



Antarctic temperature 410,000 years ago

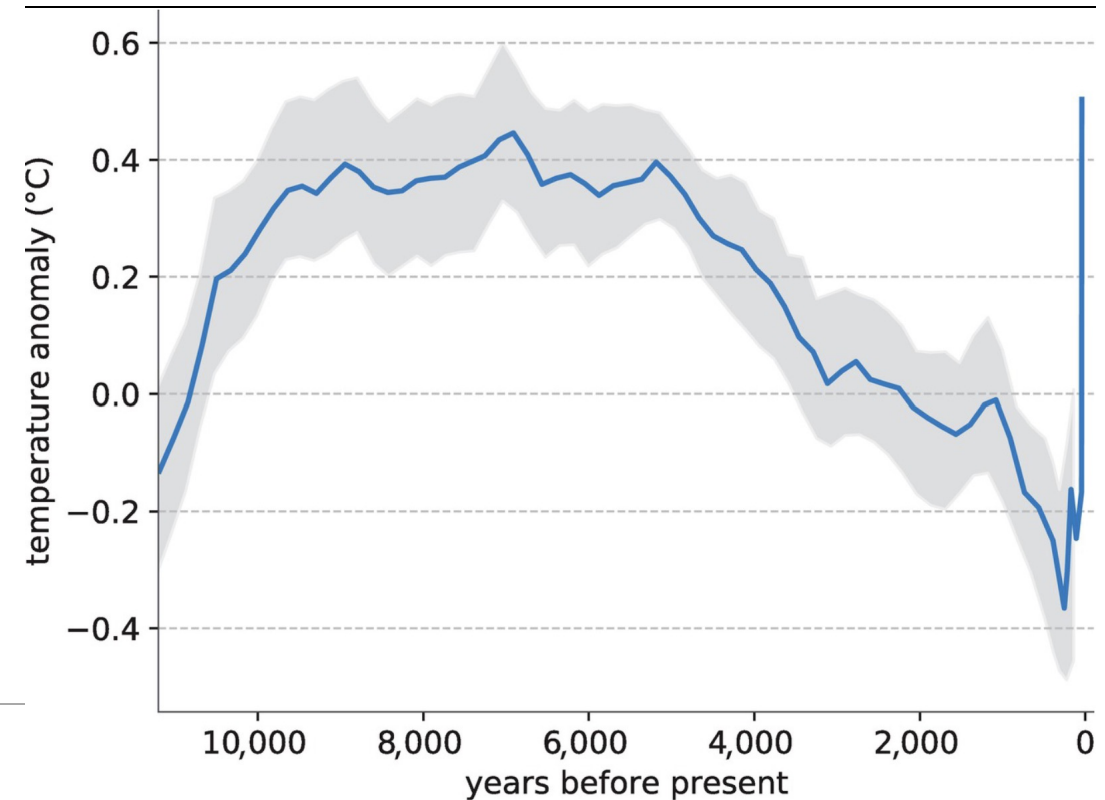
- Shape of recent ice-age cycles — slow cooling (10s of 1000s of years) whereas fast warming at the end of an ice (10,000 years).
- Overall, ice ages lasted ~ 100,000 years, while the interglacials are relatively short, lasting 10,000 to 30,000 years.
- The coldest period during an ice age, (**glacial maximum**), occurs right before the end of the ice age.
- The last glacial maximum occurred about 20,000 years ago, when the global average temperature was about 6°C cooler than modern temperatures.
- The last ice age ended about 12,000 years ago, and since then we have been enjoying a rather pleasant interglacial.



Temperature anomaly of the southern polar region (blue line) over the past 410,000 years, relative to today's temperature, constructed from an Antarctic ice core. **Carbon dioxide** (orange line) is from air bubbles trapped in the ice.

Global Temperature of the Holocene

- **Holocene** : period beginning at the end of the last ice age, about 11,700 years ago.
- Temperatures peaked about 7,000 years ago.
- Then temperature started a slow, long-term decline that bottomed out in a period 200 to 300 years ago, known as the **Little Ice Age**, during which temperatures were about 1°C below today's.
- After the Little Ice Age, temperatures began warming. Today's temperatures are comparable to peak temperatures of the mid-Holocene.



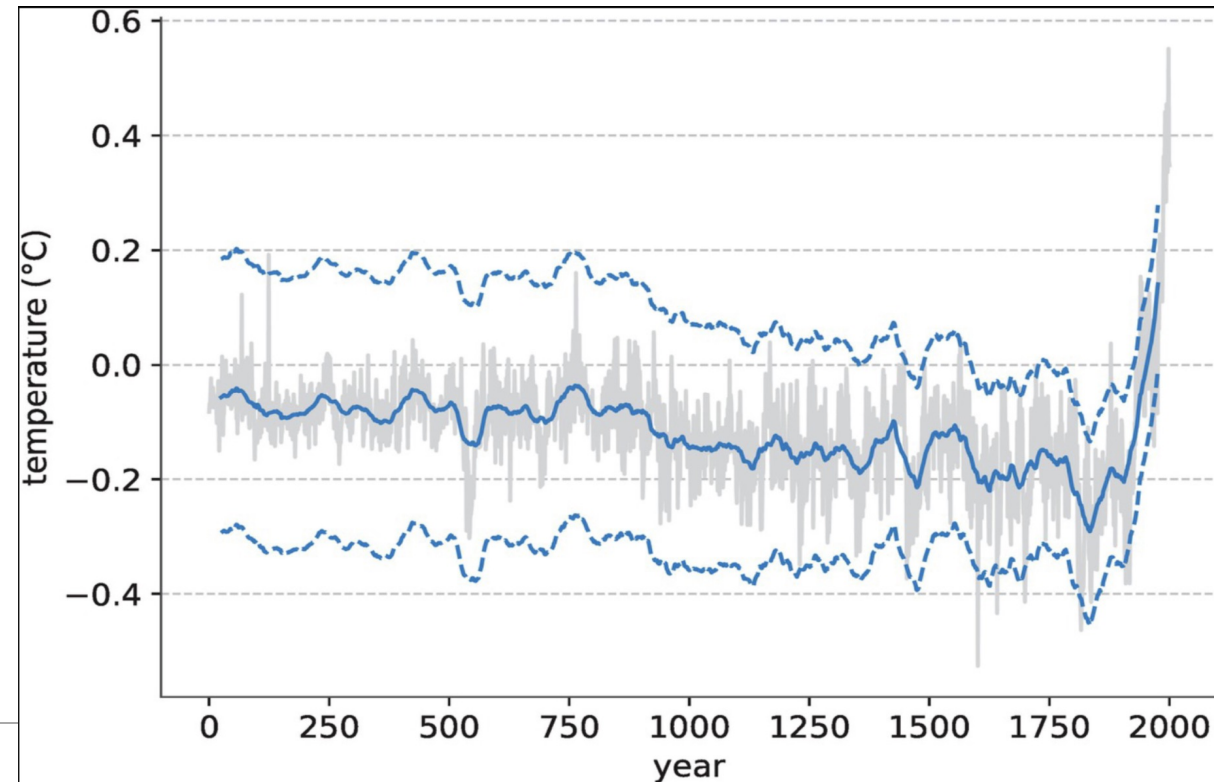
Global temperature anomaly (°C) of the last 11,000 years, based on multiple proxy records. The shaded region represents the uncertainty in the estimate. Anomalies are calculated relative to the 1961-1990 average.

Climate change from paleorecord

- Small changes in global temperature can produce major environmental changes
-  Ice ages — $\sim 6^{\circ}\text{C}$ colder than today; thick glaciers covered much of North America; global sea level was ~ 100 m lower. Major changes occurred in ecosystems and the global environment. Earth looked fundamentally different from today
-  Little Ice Age — $\sim 1^{\circ}\text{C}$ colder than today. European glaciers advanced dramatically, affecting farms and settlements. Rivers such as the Thames frequently froze. Historical records and paintings show substantially colder conditions.
-  A few degrees of global warming may sound small, but Earth's history shows that temperature changes of this magnitude can fundamentally reshape the climate, sea level, ice sheets and ecosystems.

Global average temperature over 2,000 yrs

- Temperatures generally declined over this period, until the 1800 year. At that point, we see a dramatic warming that clearly deviates from the previous 1800 years.
- Earth has been both much warmer than today and much colder.
- Looking at the more recent past (the Holocene), temperatures have been declining for about 7,000 years, until around 1800, at which time temperatures began rapidly increasing with human addition of GHGs.

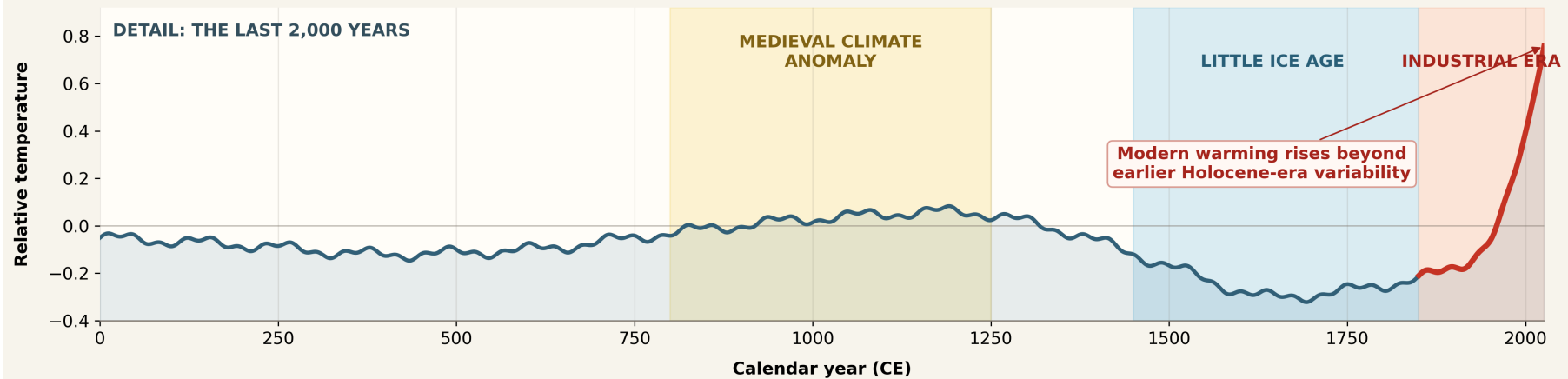
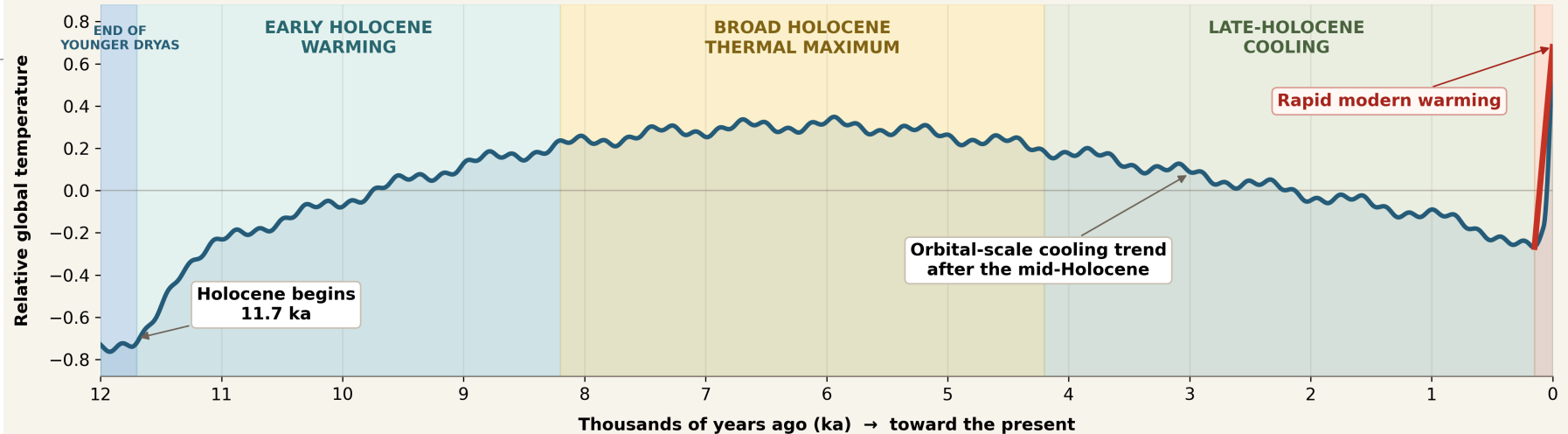


Global average temperature anomaly between years 0 AD to 2000 AD. Gray line shows the annual average values, while the blue line shows 50-year smoothed values. The dashed blue lines show the 10-90 percent confidence interval of the smoothed value. Anomalies are calculated relative to the 1951-1980 average. Data are ensemble-average 2-m air temperature (that is, temperature 2 m above ground)

Climate history last twelve thousand Years

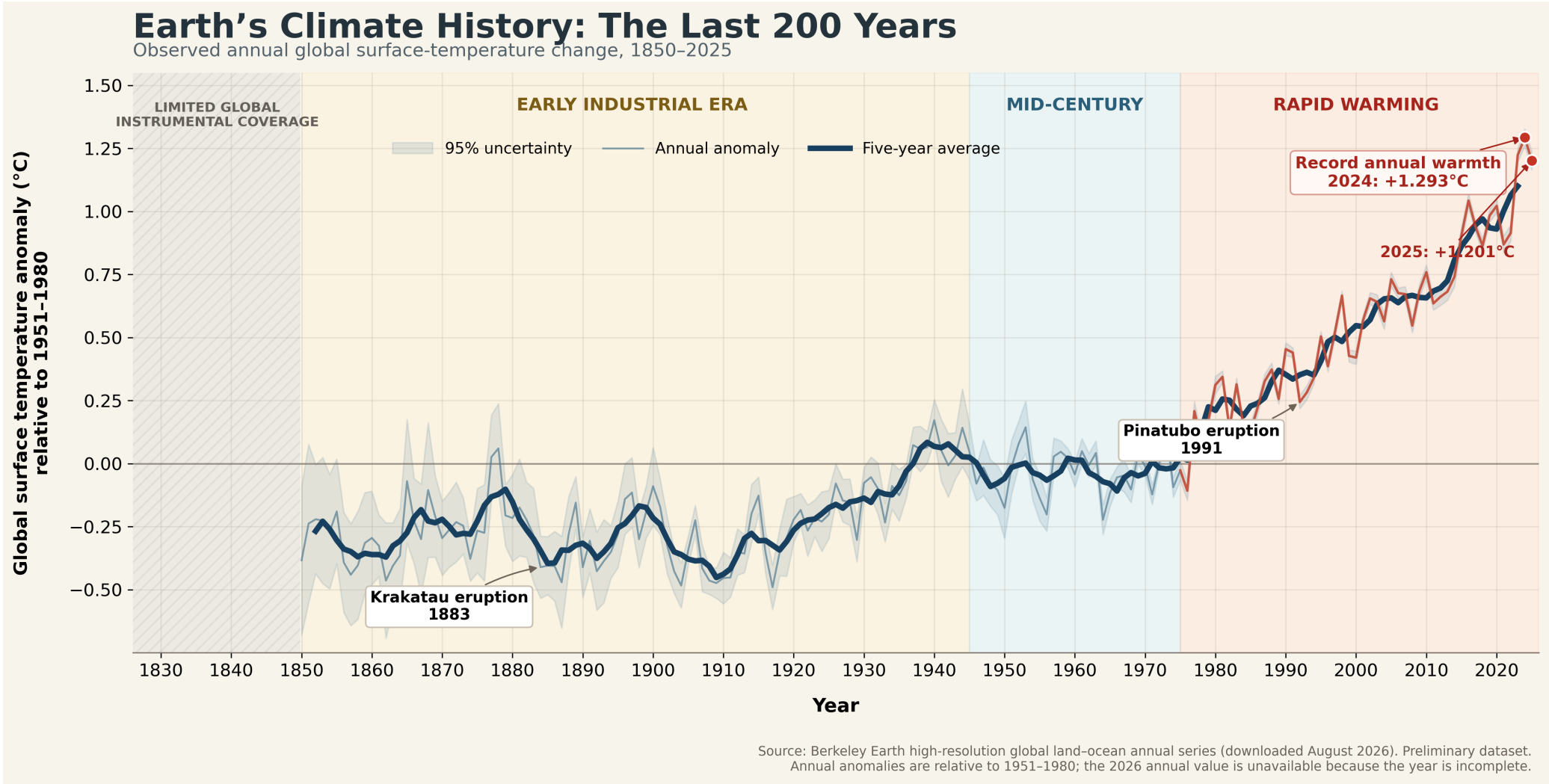
Earth's Climate History: 12,000 Years Ago to the Present

The Holocene was relatively stable, but regional and global variations occurred before rapid industrial-era warming



Schematic synthesis of multi-proxy Holocene reconstructions and the instrumental-era trend. Relative values emphasize timing and direction; regional expressions—especially the Holocene Thermal Maximum and Medieval Climate Anomaly—were not globally synchronous or uniform.

Climate history last two thousand Years



Today's Conditions

- Humans have been around for a century or two. This means that the range of global average temperatures our society has experienced is small, around 1°C .
- As our climate continues to warm, we are departing from conditions under which human society developed and thrived.
- More troubling, the warming we are experiencing is extraordinarily rapid.
- For example, the warming over the past century (approximately 1°C in about a century) is around 16 times faster than the average rate of warming coming out of the last ice age (roughly 6°C in 10,000 years corresponds to an average warming of $0.06^{\circ}\text{C}/\text{century}$).

Summary

- Most well-studied and reliable source of temperature data for the past century is the surface thermometer network. It shows a global and annual average warming of 1.1°C between the late 19th century and the 2010s.
- These are further confirmed by satellite measurements of temperature, amount of ice on the planet, ocean heat measurements, and sea-level measurements.
- Because of the overwhelming evidence supporting it, the scientific community has concluded that the observed warming of the climate system is beyond doubt — the IPCC uses the word “unequivocal.”
- Earth’s climate has varied widely over its 4.5-billion-year history. The geologic record shows that the climate has been both warmer and cooler than today’s climate.
- Over the past few million years, the Earth has oscillated between ice ages and warmer interglacial periods. Ice ages are about 6°C cooler than the interglacials. The Earth is currently in an interglacial.
- We are currently in the Holocene, the period that began at the end of the last ice age. Temperatures peaked about 7,000 years ago and declined until about 200 years ago, after which temperatures began rapidly rising. This abrupt rise coincided with the beginning of society-wide combustion of fossil fuels.